

“Composition Dependent Modulus In Fe-Ni System”

Submitted by

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Entry number: 2020MS10742

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A report submitted on partial fulfillment of the requirement for the degree of Bachelor of Technology in Materials Engineering in the

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Academic year **2023-2024**

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Table.4 Showing the hardness and modulus values across the Ni- S.S. interdiffusion zone

Table.5 Hardness and modulus at the three points in the Ti-Ni region

3. Nomenclature

EDXS – Energy Dispersive X-ray Spectroscopy

SEM – Scanning Electron Microscopy

S.S. - Stainless Steel

4. Abstract

The diffusion couple technique was used to develop a methodology that can be used to study the correlation between mechanical properties such as hardness and modulus with composition. Variation of hardness with composition and modulus with composition is studied using the developed methodology in the Fe – Ni system. For this purpose, Ni-S.S. and Ti-Ni-S.S. assemblies were prepared and then diffused using the diffusion couple technique. The nanoindentation test is conducted at various points across the interdiffusion zone to obtain hardness and modulus values. EDXS is performed for compositional analysis across the interdiffusion zone.

5. Introduction

A diffusion couple comprises two pieces of metals kept in close contact, which allows thermal interdiffusion and creates composition gradients and intermetallic compounds [1]. The diffusion couple technique is better suited to study the mechanical properties of various alloys, such as modulus and hardness, than the conventional method. The conventional method is the one-alloy-at-a-time approach. Diffusion couple eliminates the need for making individual alloys to study the mechanical properties. The composition map of the alloys formed along the interdiffusion zone can be obtained by the Energy Dispersive X-ray Spectroscopy technique (EDXS). The Nano-Indentation hardness test can be performed along the interdiffusion zone to get the modulus and hardness at localized spaces. The current study will mainly focus on developing Ni-S.S. alloy and Ni-S.S.-Ti alloy through diffusion couple techniques to obtain the variation of hardness and modulus correlation with composition along the interdiffusion zone.

This study is useful to develop a methodology consisting of a single experiment to analyze the variation of hardness and modulus along the interdiffusion zone. S.S. was chosen for this study because of its various engineering applications. This technique can be used to predict/check/verify the usability of an alloy of two metals for a particular engineering application on the basis of the alloy's mechanical property. Thus the trends obtained from this study will be useful to understand about an alloy of S.S. with other metal, which can then be used in an engineering application.

6 Literature Review

6.1 Parabolic Growth Law of Diffusion

In a diffusion-controlled process, the rate of increase in layer thickness is inversely proportional to the thickness when only a single layer grows.[2]

$$\frac{dx}{dt} = \frac{1}{x}$$

where x is the layer thickness and t is time.

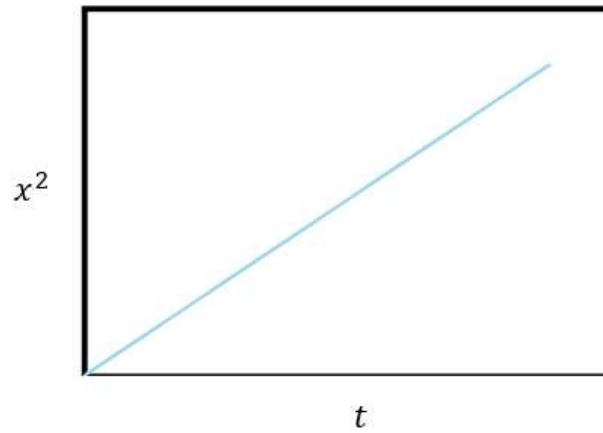


Fig1. Showing dependence of time with layer thickness.

From the figure, it can be shown that increasing time increases the length of the interdiffusion zone thus this law will become a basis of selecting holding time based on which we can get an optimized holding time.

7. Description of Experimental Procedure

7.1. Simulation

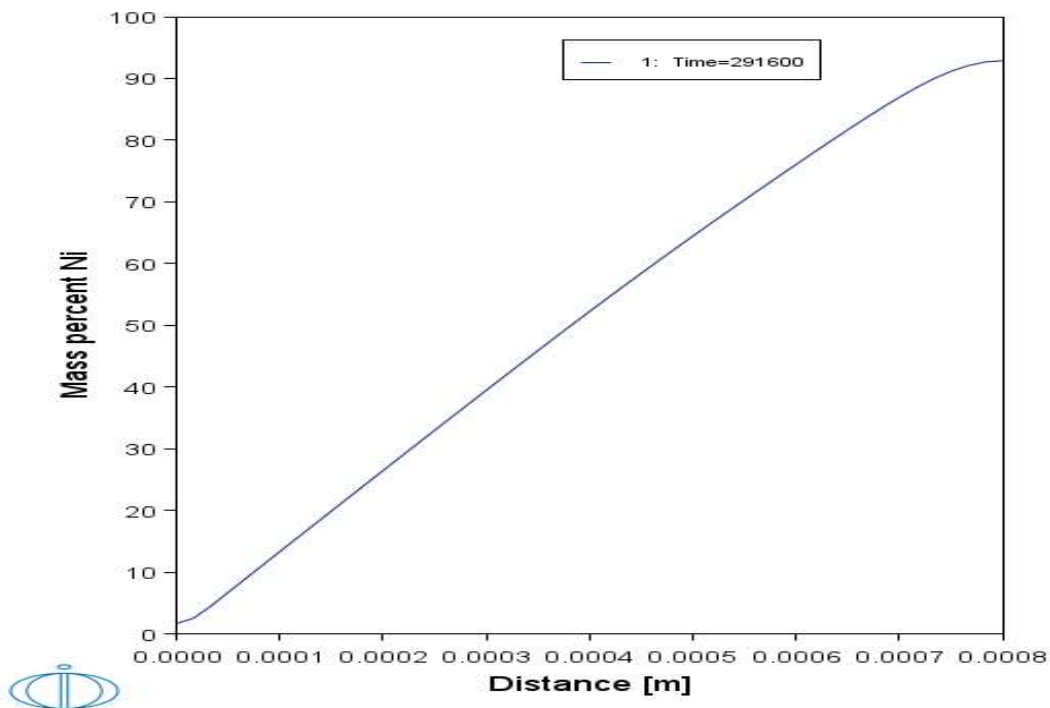


Fig.2 Showing composition of interdiffusion zone of nickel and iron

The optimized temperature and optimized holding time is necessary for this technique, as it would lead to an optimized length of the interdiffusion zone. This is necessary to get the correct measure of a mechanical property. To get the optimized temperature and holding time simulation was used.

Simulation of diffusion of Iron in Nickel was carried out using Thermo-Calc Educational 2023b software at several temperatures and holding time to optimize the holding time and temperature for diffusion. Optimized holding time = 100 hours. Optimized temperature 1100 °C.

7.2. Material and pre-treatment

For the study, the dimensions of nickel and titanium samples were chosen as 0.3mm*0.6mm*1mm. Dimensions of the 316L stainless steel sample were chosen as 03mm*0.6mm*2mm. Several nickel, titanium, and stainless-steel samples were cut into each desired dimension using the slow-speed and high-speed cutter machines. The samples were then polished using a 600-grit sandpaper wheel and examined under an optical microscope. This process continued until any grooves larger than the size of the sandpaper particles were eliminated. This procedure was repeated with 800, 1200, 2000, and 4000 grit.

7.3. Experiment and Characterization of material

For the preliminary test an assembly of Ni-SS was prepared and were heat-treated under vacuum conditions to carry out diffusion bonding in TEMPSENS TF-T-1200 at 1100 °C for 25 hours.

The bonded assembly was polished on 600, 800, 1200, 2000 and 4000-grit sandpaper followed by diamond paste and colloidal silica. The composition map of the metallographically bonded assembly was obtained by SEM-based energy dispersive spectroscopy (SEM-EDS and micrographs). Nanoindentation test using berkovich indenter and 10mn load and Vickers hardness test was then conducted along the interdiffusion zone of the assembly, hardness and modulus values were recorded.

Two bonded assemblies of SS-Ni-Ti and Ni-SS-Ti were prepared respectively and were heat-treated under vacuum conditions to carry out diffusion bonding in TEMPSENS TF-T-1200 at 1100 °C for 100 hours. Similarly, their composition map and hardness values were recorded.

The bonded assemblies were polished on 600, 800, 1200, 2000 and 4000-grit sandpaper followed by diamond paste and colloidal silica. The composition maps of the metallographically bonded assemblies were obtained by SEM-based energy dispersive spectroscopy (SEM-EDS and micrographs). Nanoindentation and Vickers hardness test were then conducted along the two interdiffusion zones of each assembly, hardness and modulus values were recorded.

8. Results and Discussion

8.1 Composition Analysis of S.S.-Ni Assembly Annealed for 25 hours

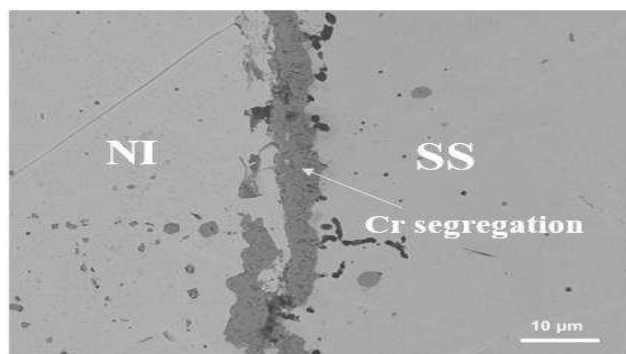


Fig.3 Showing Microstructure of Ni-S.S. sample across the interdiffusion zone

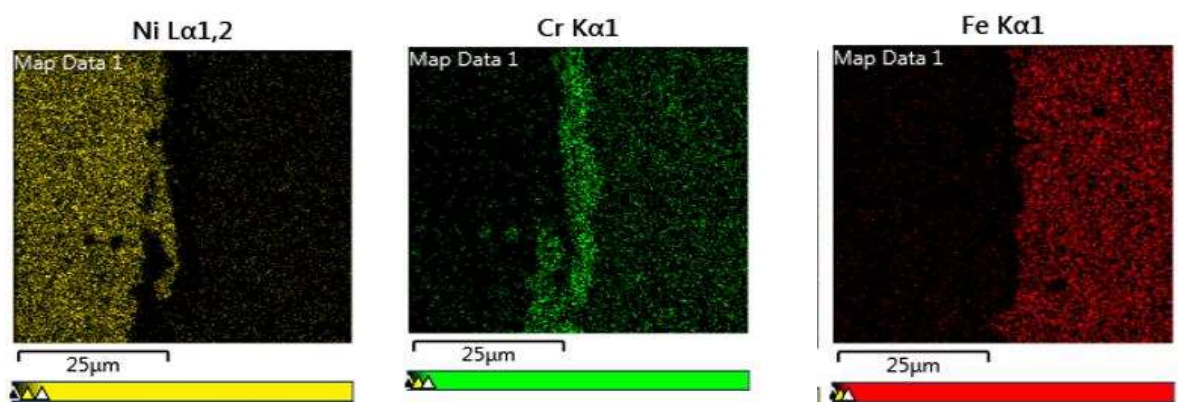


Fig.4 Showing the compositional map of Fe, Ni, Cr, across the interdiffusion zone

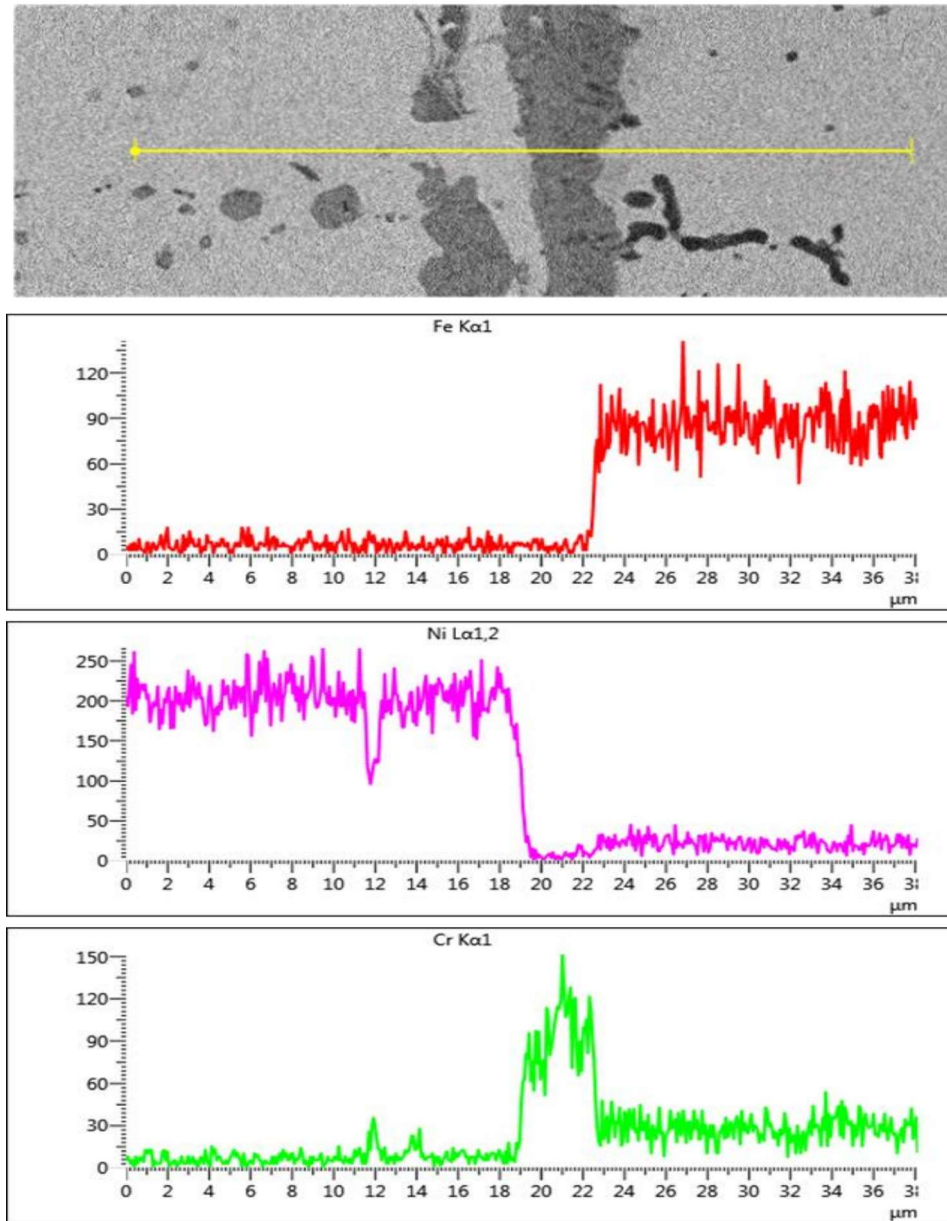


Fig.5 Showing the line scan across interdiffusion zone

Initial examination of the Ni-S.S. assembly showed that it appears to be well-joined. Composition analysis of diffused assembly of Ni-S.S. using S.E.M was done and three main regions were identified- i) Nickel rich region ii) Chromium rich region and iii) Iron rich region. Inter-diffusion zone was found to be a chromium rich region. Chromium from 316L stainless steel got segregated into the interdiffusion zone. There was not significant penetration of Iron into nickel and vice-versa. The interdiffusion zone identified from the line scan was of length 4-6 micrometres. There was no significant formation. A point scan at the interdiffusion zone

showed the presence of oxygen apart from chromium. Smaller penetration of iron into nickel can be due to a shorter holding time during annealing. A proper 'S' curve in diffusion of Iron in Nickel was obtained for 81 hours using the Thermo-Calc software. The preliminary study showed that the nickel stainless steel can be joined in 25 hours, it also verified the absence of significant penetration for this duration as predicted by the simulation.

8.2 Hardness Tests of S.S.-Ni Assembly Annealed for 25 hours

Nanoindentation test was performed on the assembly at many regions to sample all the three different regions and hardness values were recorded.

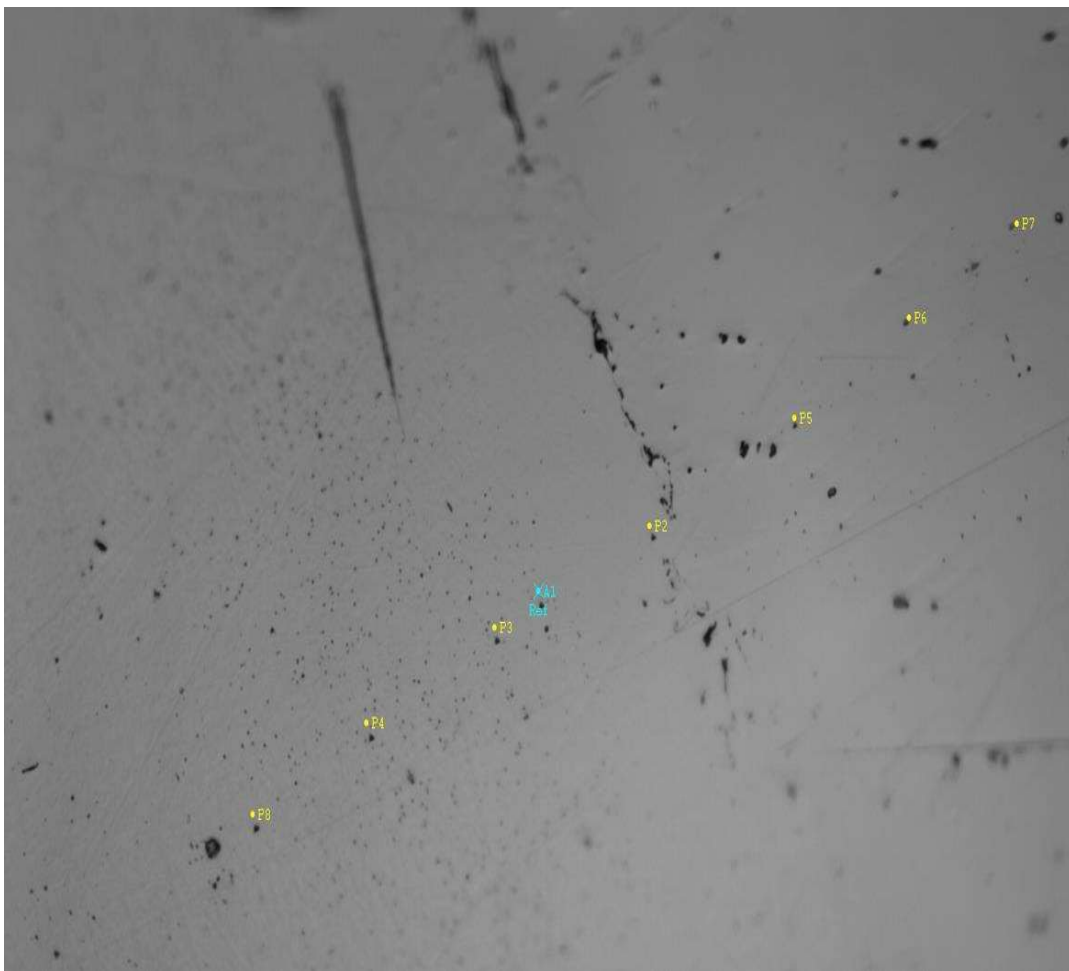


Fig.6 Showing seven different indents taken for hardness across the Interdiffusion zone.

S.No.	Hardness in Hv	Modulus in GPa
1	279.1	230.8
2	313.1	355.7
3	294.5	217.8
4	271.2	282.5
5	328.8	245.9
6	317.1	251.5
7	246.2	374.5

Table1 Showing the nanoindentation hardness and hardness values, 2,3 and 7 corresponds to Ni rich region where as 4,5 and 6 corresponds to S.S. rich region, 1 corresponds to Cr rich region

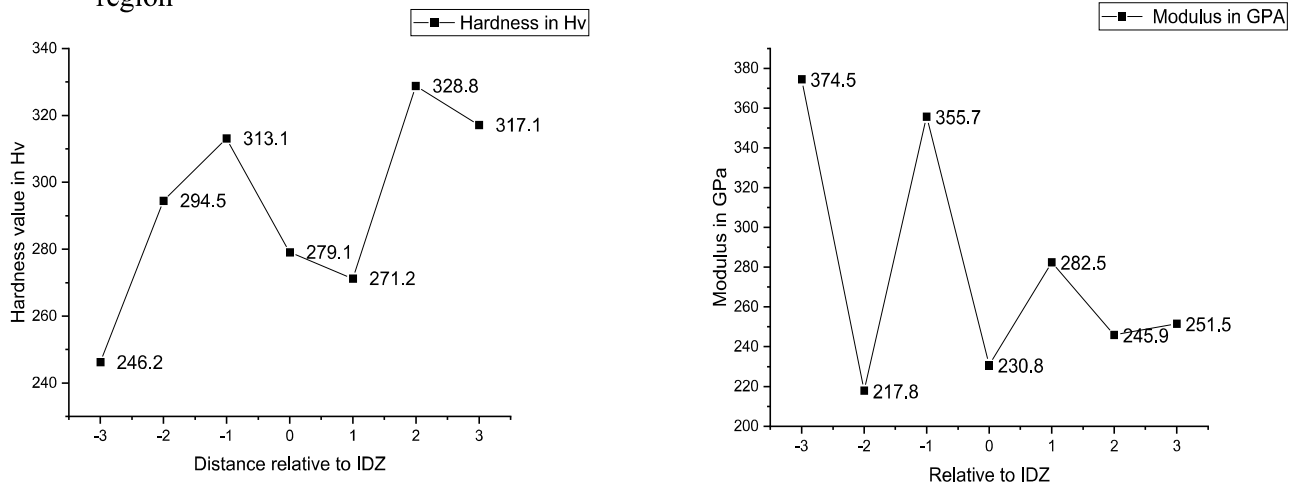


Fig.7 Showing Hardness and modulus values with distance relative to IDZ, -3 -2 -1 are in Ni rich region, 1,2,3 are in S.S. rich region 0 is in chromium rich IDZ.

The average hardness in Ni, S.S. and Cr rich region is found to be 284.6 Hv, 305.6 Hv and 279.1 Hv respectively. This corresponds to the fact that Stainless steel is harder than Ni followed by Cr. The hardness value is minimum. The variation of modulus is non-uniform. Average modulus in Ni, S.S. and Cr rich region is found to be 316 GPa, 259.96 GPa and 230.8 GPa. The modulus values

obtained for Ni and S.S. are slightly greater than the actual value, this can be because of annealing of sample.

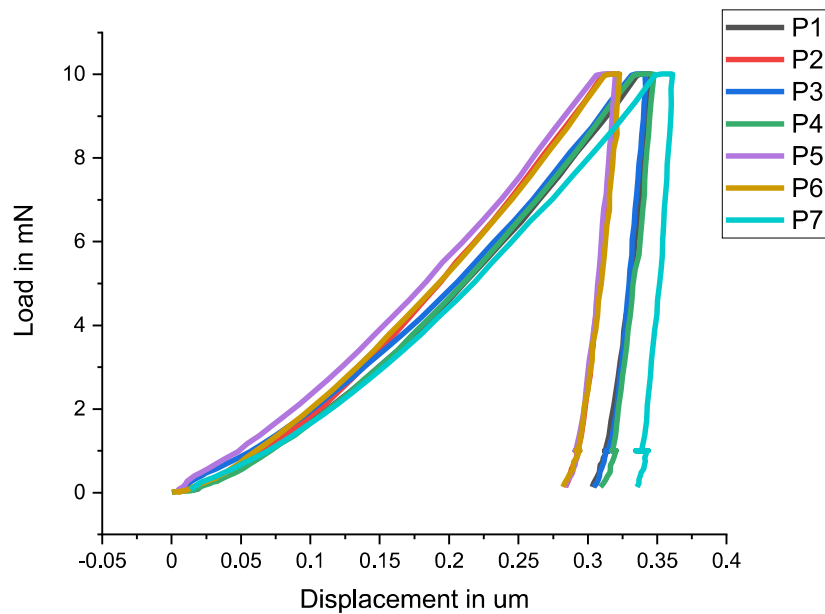


Fig.8 Showing the P-H curve formed by the nanoindentation test of all the points

Vicker's hardness test was performed on the assembly and results were recorded. This test was performed by applying a 980.1 mN load. The average value in the Ni, S.S. and Cr rich region was found to be 104.33Hv, 136Hv and 152Hv respectively. These values are smaller than the nanoindentation values. Generally the nanoindentation hardness is 10-30% of vicker's [3]. The vicker's hardness test is not localized around the interdiffusion zone. The length of the interdiffusion zone as obtained from the line scan is less than 10 micrometres whereas the size of the indent in a vicker's hardness test has a diameter of around 30 micrometres.

S.No.	Hardness in Hv
1	136
2	135
3	137
4	152
5	107
6	102
7	104

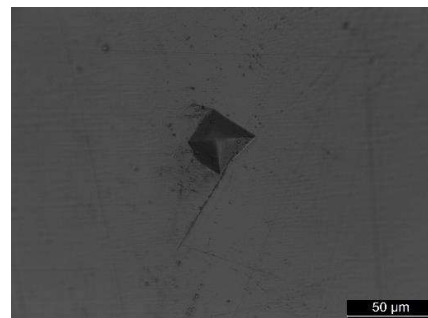


Table.2 Shows the hardness value obtained at different points by vicker's hardness test.

Fig.9 Showing the indentation created by Vickers indenter, around 30 um in diameter.

8.3 Composition Analysis of Ni-S.S-Ti Assembly Annealed for 100 hours

On the preliminary examination of this assembly, it was found that Ti did not join with S.S. This can be due to formation of various intermetallic compound like Fe_2Ti , FeTi , χ phase, Cr_2Ti and $\text{Fe}_2\text{Ti}_4\text{O}$ in the diffusion interface. This leads to embrittlement of the diffusion joint. Also, the large difference between the linear thermal coefficients of Ti and S.S. leads to high internal stress between the materials [4-7]. Hence intermediate materials should be used to study the diffusion between S.S. and Ti. However, studying Ni-S.S. assembly was found to be important because increasing the holding time will also increase the performance of diffusion. The penetration of Ni to S.S. and S.S. to Ni should be significant and a 'S' curve should be obtained. Therefore, discarding the Ti, the study was continued with Ni-S.S. assembly formed by diffusion at 1100 °C for 100 hours. Hereon in this document, this assembly will be called Ni-S.S. assembly.

8.4 Composition Analysis of Ni-S.S. Assembly Annealed for 100 hours

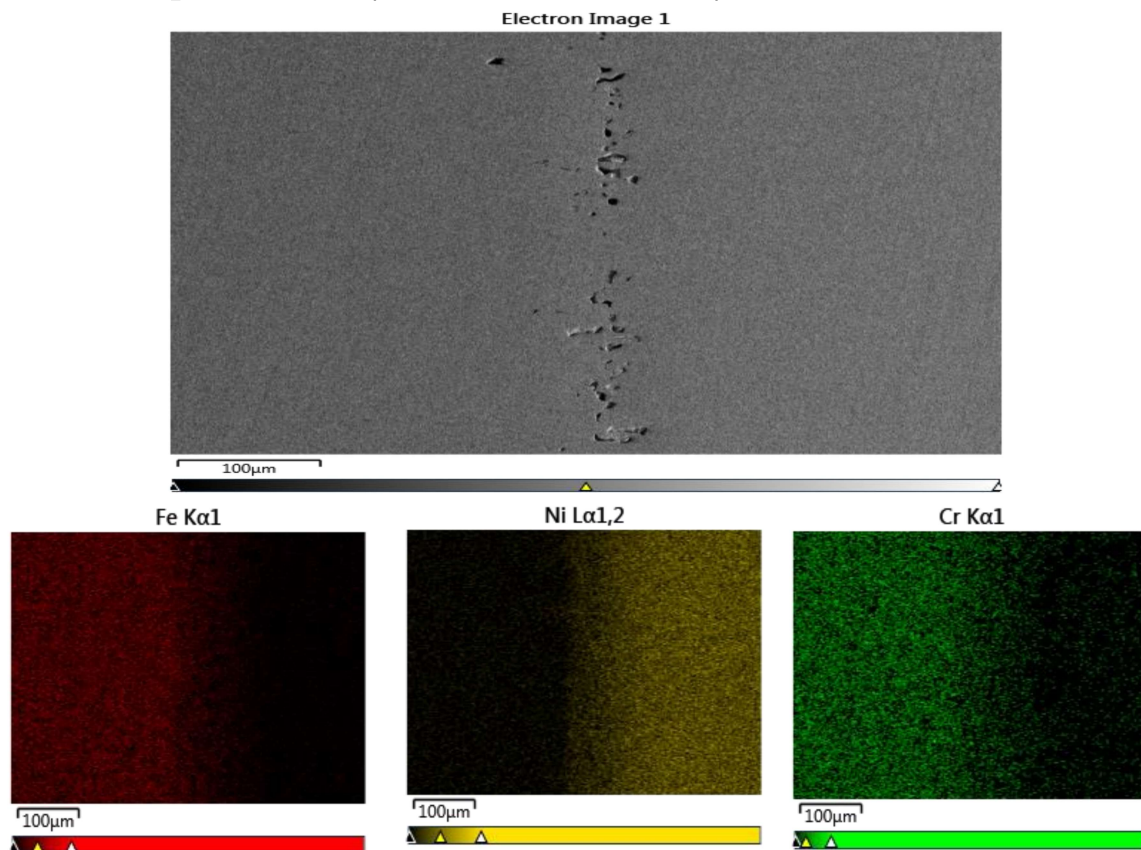


Fig.10 Showing the microstructure and compositional map of Fe, Ni and Cr along the interdiffusion zone

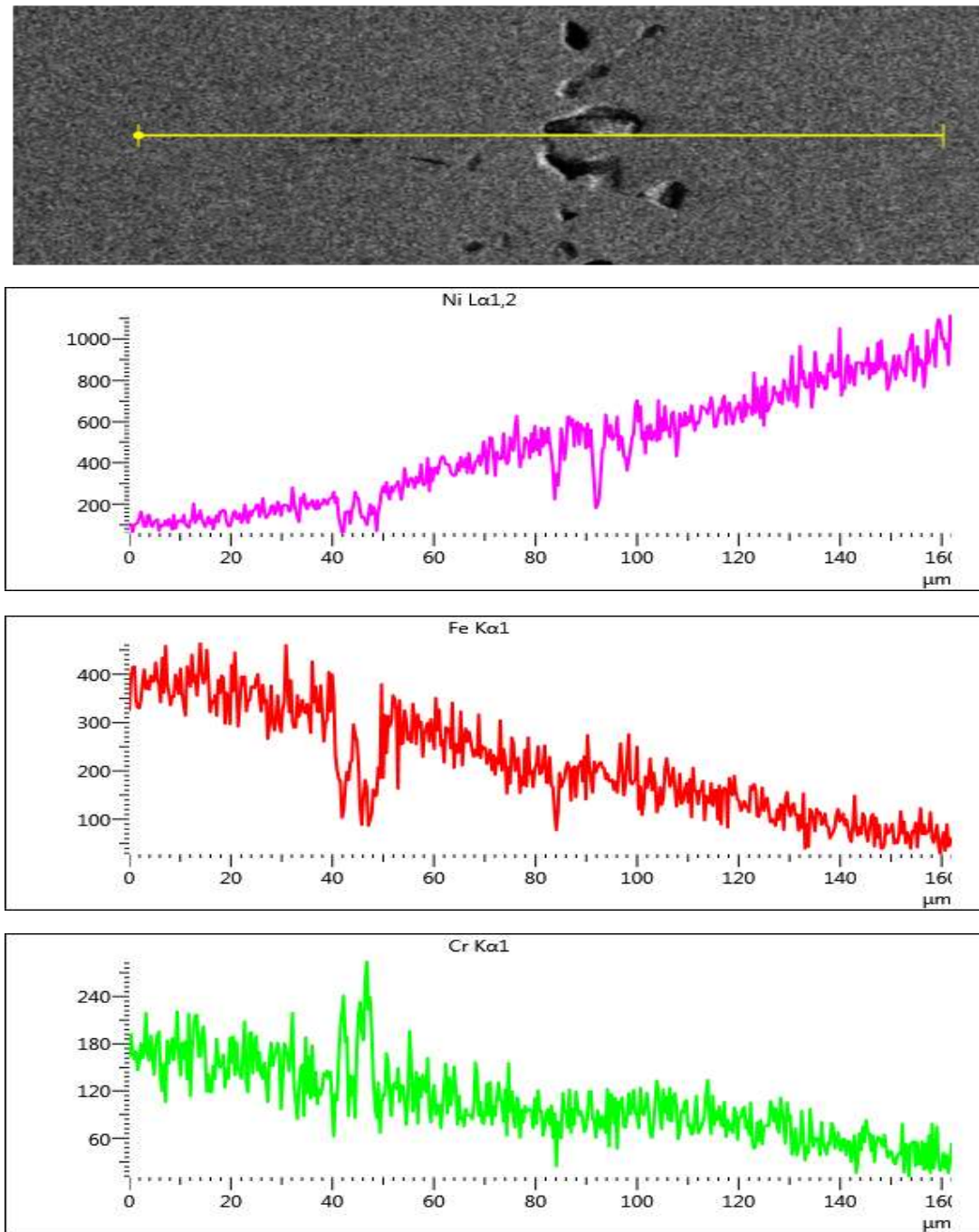


Fig.11 Showing line scan along the interdiffusion zone of Fe, Ni and Cr

Compositional analysis of the assembly showed presence of two major region- i) Fe rich region ii) Ni rich region. There is no region which is only chromium rich. Chromium is present mainly in Fe rich region, with some penetration in Ni rich region. From the

composition map, it can be noticed that the penetration of Cr is better than the penetration of Fe into Ni. Line scan showed a gradual change in composition of Fe and Ni across the interdiffusion zone. The width of the interdiffusion zone is around 150 micrometres.

8.5 Hardness Test of Ni – S.S. Assembly Annealed for 100 hours

Nanoindentation test was performed on the assembly at many regions to sample all the different regions and hardness values were recorded.

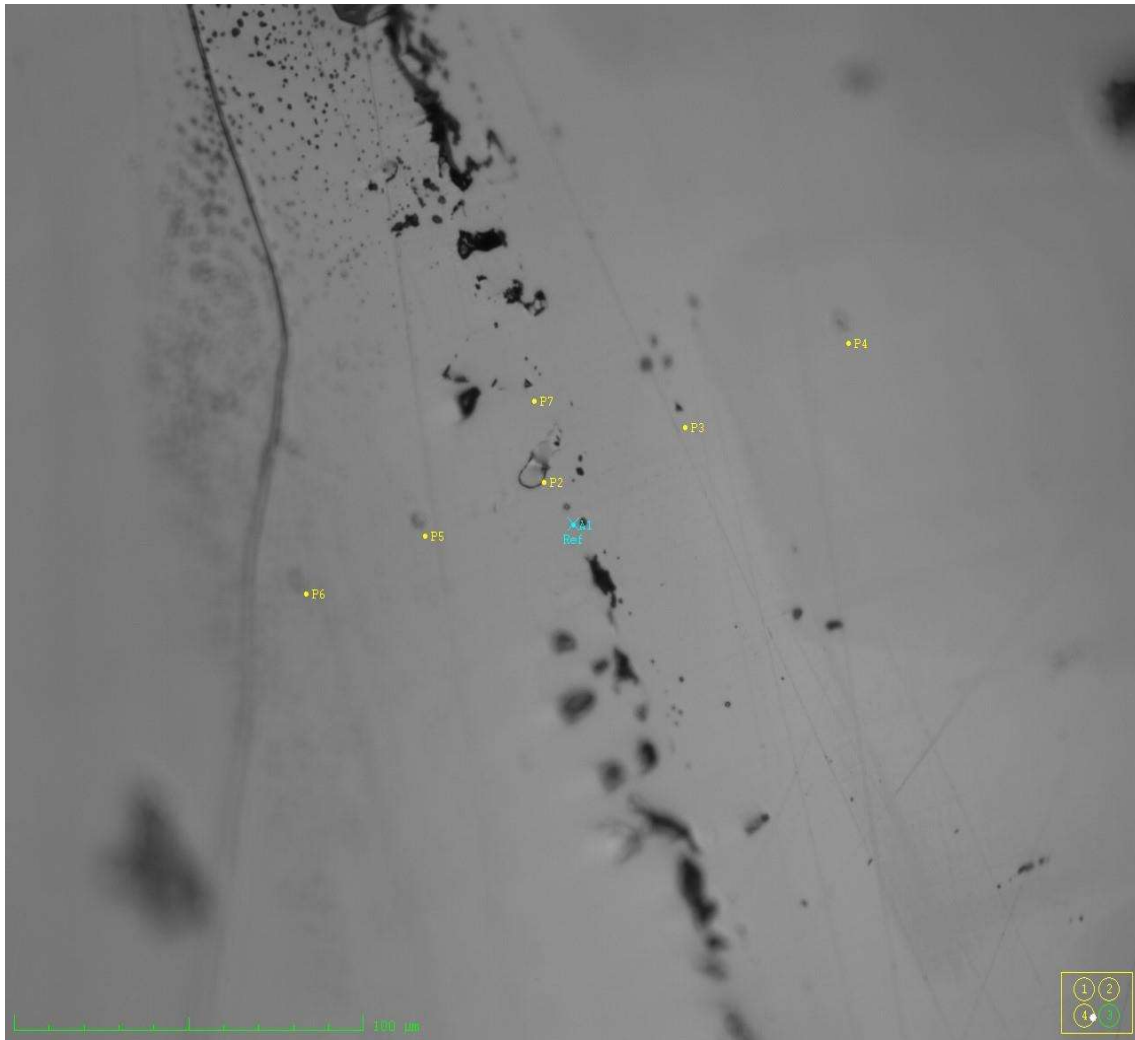


Fig.12 Showing different indents taken over the sample by the nanoindentation machine to get hardness.

S.No.	Hardness (Hv)	Modulus in (GPa)
1	393.5	222.9
2	351.4	206.7
3	356.8	200.4
4	248.2	214.9
5	481.8	200.2

Table3 Showing hardness and modulus value across the interdiffusion zone

An increased holding time has hardened the sample. The average hardness of this assembly is 366.34 Hv, while the average hardness of previous assembly was 292.85 Hv. There is no significant variation across the interdiffusion zone is found in modulus value. The average modulus is 209.02 Hv. The range of modulus on these 5 points across the interdiffusion zone is also only 20 Hv.

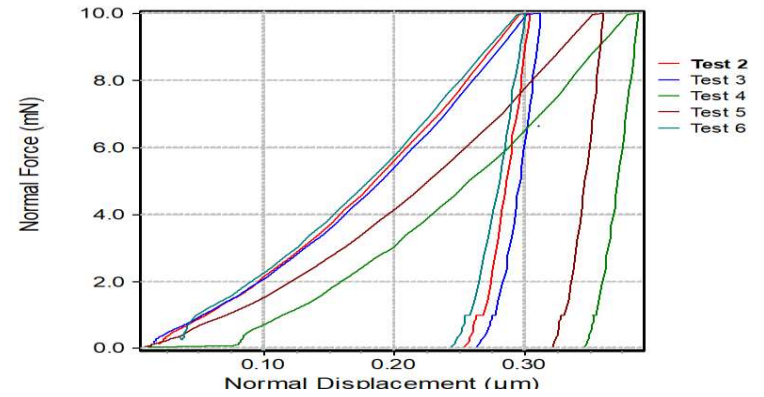


Fig.13 Showing P-H curves at points

8.6 Composition Analysis Of S.S.-Ni-Ti Assembly Annealed for 100 hours

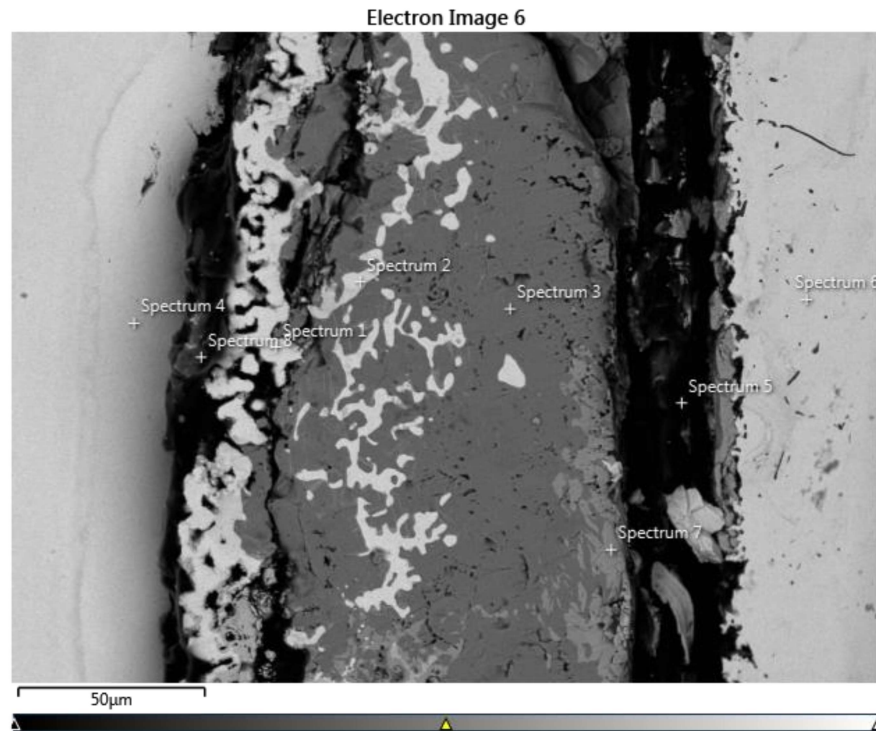


Fig.14 Showing the Microstructure of the S.S-Ni interdiffusion zone.

On preliminary examination, it was found that the assembly is well-joined; across the S.S-Ni interdiffusion zone many different compounds were formed. The above figure shows the

microstructure. Point scan on 8 different points were performed to analyse the compositions at different points.

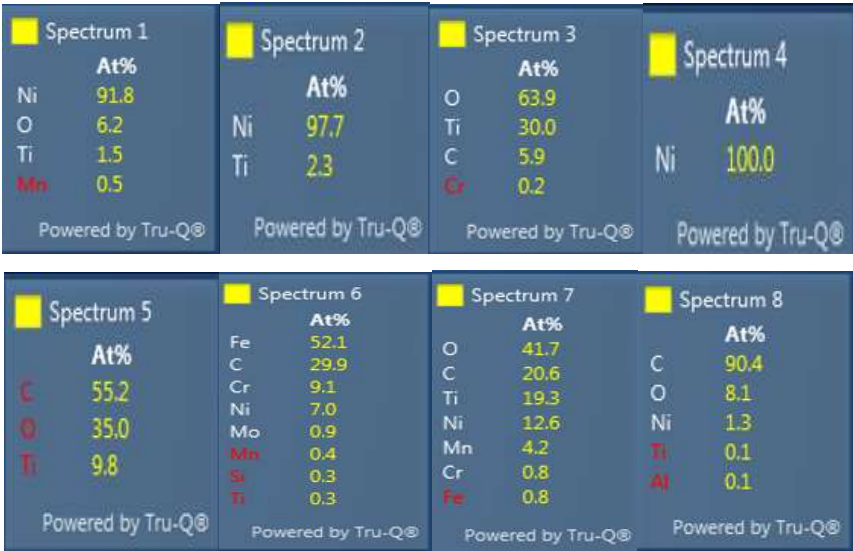
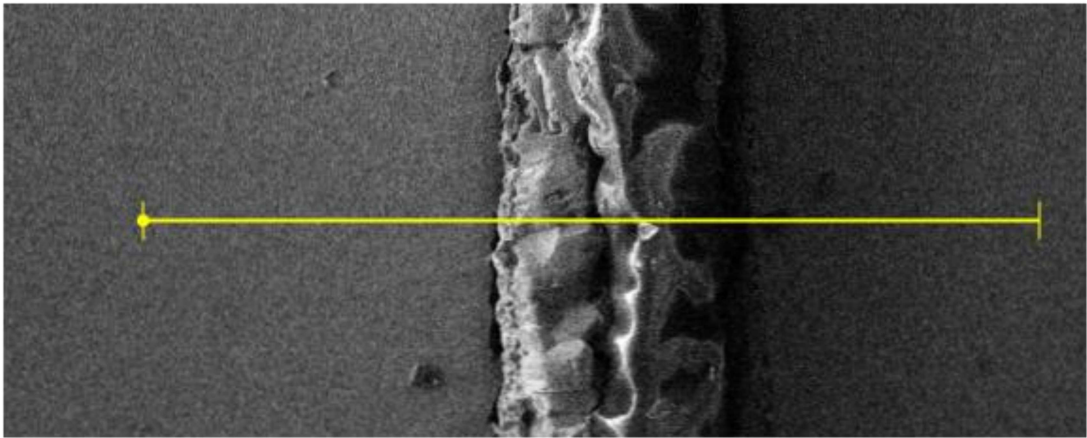


Fig.15 Showing the results of the point scan at 8 different points across the interdiffusion zone.

Points 1, 2, and 4 are found to be nickel-rich, but trace amounts of titanium are also found. Significant amounts of titanium are also found in point 3. Thus, titanium has also diffused throughout the Nickel region to reach this diffusion zone. Iron is mainly found only in points 6 and 7. Points 5 and 8 are carbon and oxygen-rich, mainly outliers as oxygen should not be present in the assembly after diffusing it in a vacuum. It can be due to the trapping of ethanol during etching in the cracks of the joint.



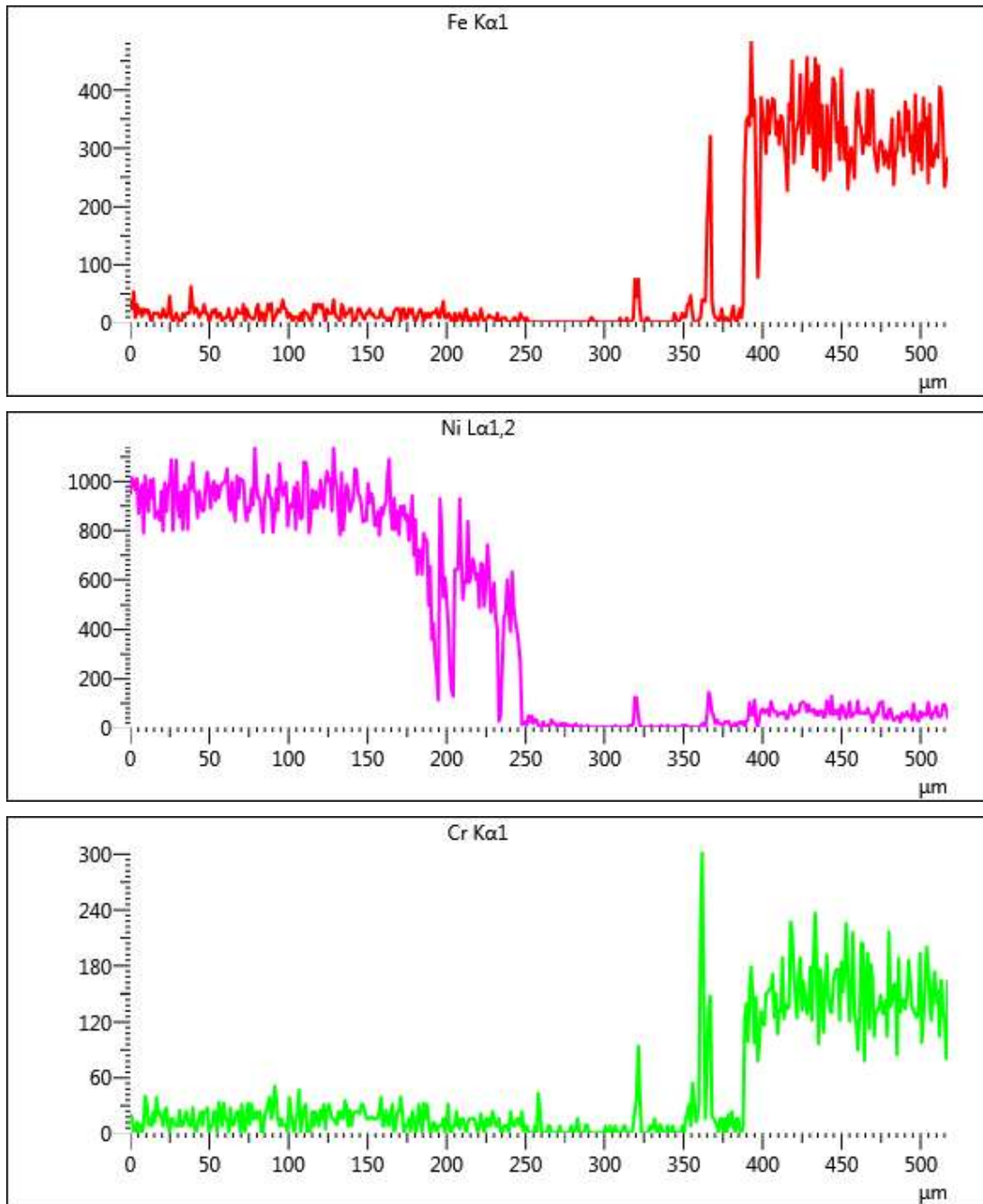


Fig.16 Showing the line scan for Fe, Ni, and Cr across the interdiffusion zone of S.S-Ni

The lines scan across the interdiffusion zone show that there is no significant diffusion between S.S. and Ni across this line. The line and point scans showed that the interdiffusion zone has cracks across this line.

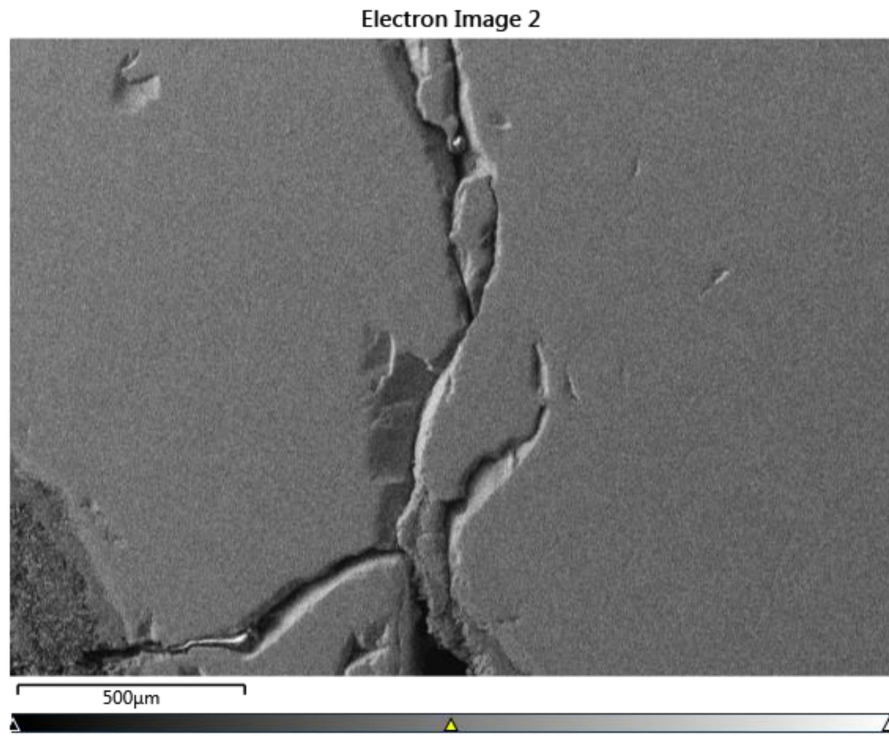


Fig.17 Showing microstructure across Ni-Ti interdiffusion zone

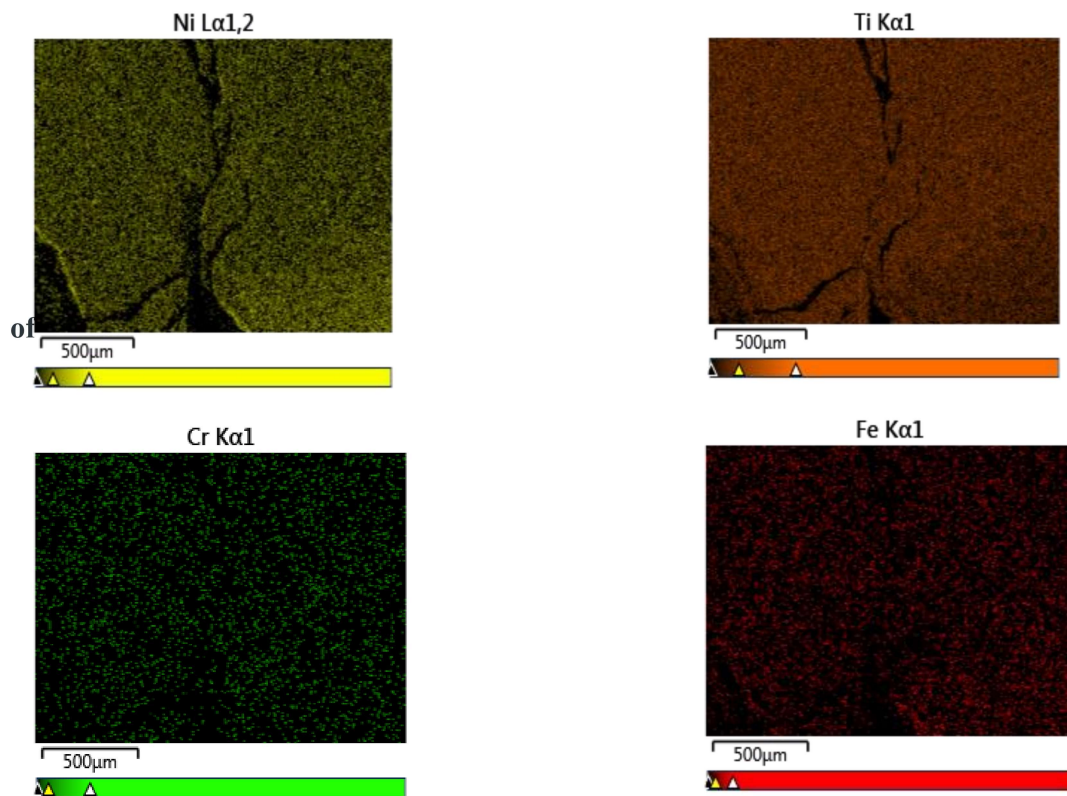


Fig.18 Showing compositional map of Ni, Ti, Cr, and Fe in Ti-Ni region

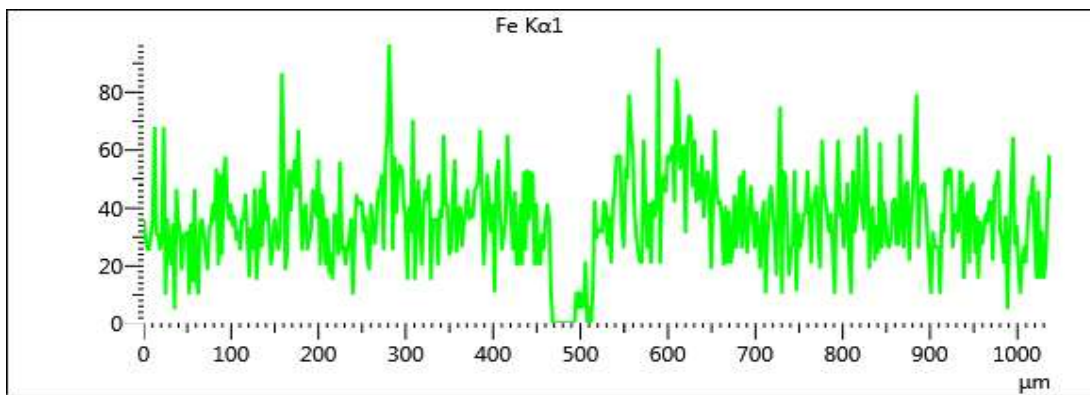
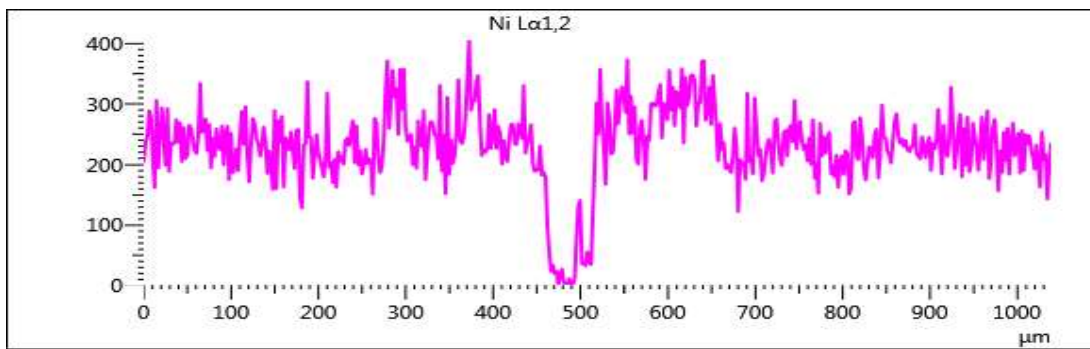
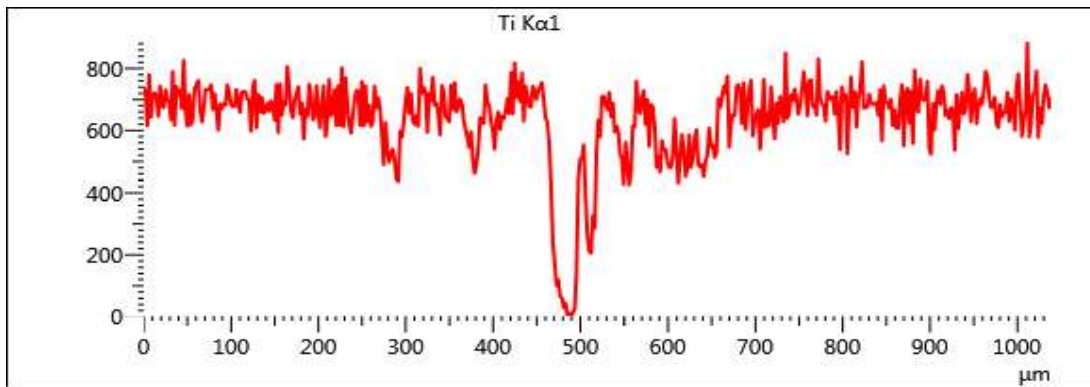
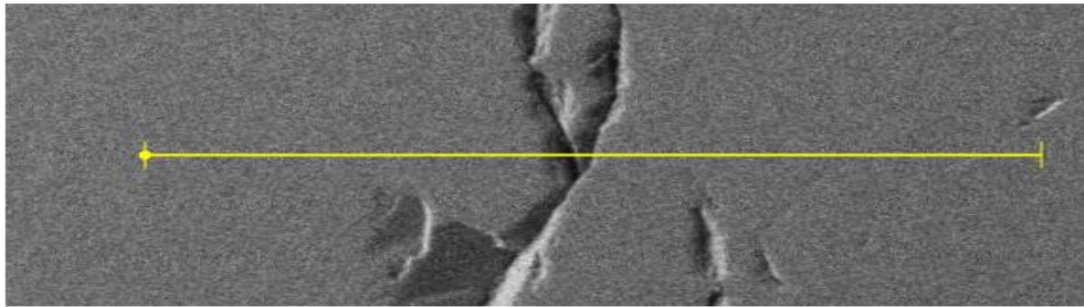


Fig.19 Showing line scan of Fe, Ni, and Ti across the Ti-Ni region

The diffusion between Ni and Ti is uniform. Quantities of Fe are also found in the region. There is no evidence of phase formation of a particular alloy. Cracks are present across the interdiffusion zone.

8.7 Hardness of S.S.-Ni-Ti Alloy Assembly Annealed for 100 hours

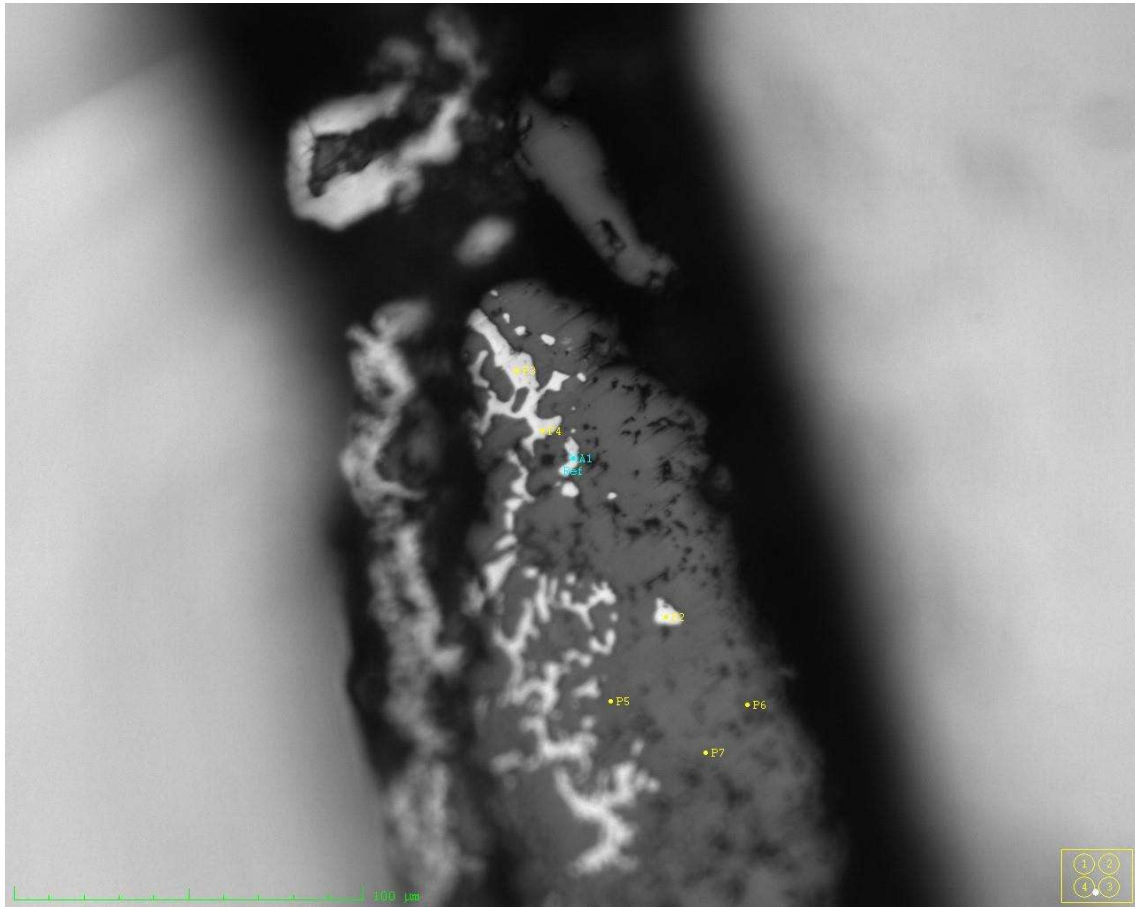


Fig.20 Showing the indents taken across the Ni-S.S. interdiffusion zone.

S.No.	Hardness Hv	Modulus Gpa
1	469.2	214.4
2	502.3	241.8
3	439.2	220.2
4	425.4	223.3
5	461.9	199.0
6	517	224.3

Table.4 Showing the hardness and modulus values across the Ni- S.S. interdiffusion zone

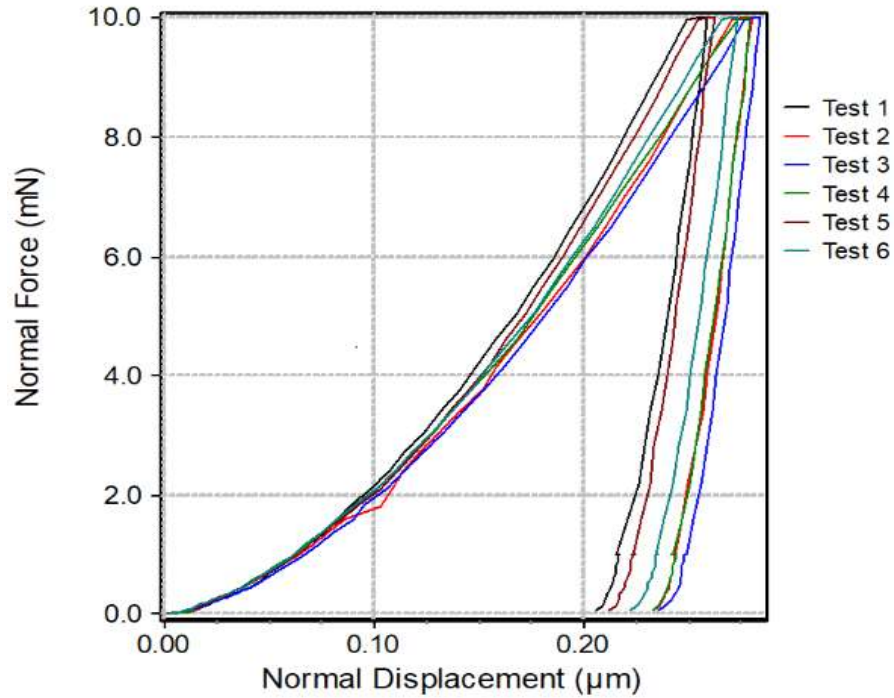


Fig.21 P-H curve of the points across Ni-S.S. interdiffusion zone

The average hardness across the Ni-S.S. interdiffusion zone is 469.1 Hv. The average modulus across this interdiffusion zone is 220.5 Gpa. Points 2, 5, and 6 are in Titanium and Iron rich regions. Hence, this region's average hardness and modulus are 493 Hv and 221.7 Gpa, respectively. Points 1,3 and 4 are predominantly Ni-rich. This region's average hardness and modulus value are 444.6 Hv 219.3, respectively. The average hardness value for this assembly is greater than that of the Ni-S.S. assembly diffused for 100 hours because of the significant diffusion of harder metal titanium in this assembly.

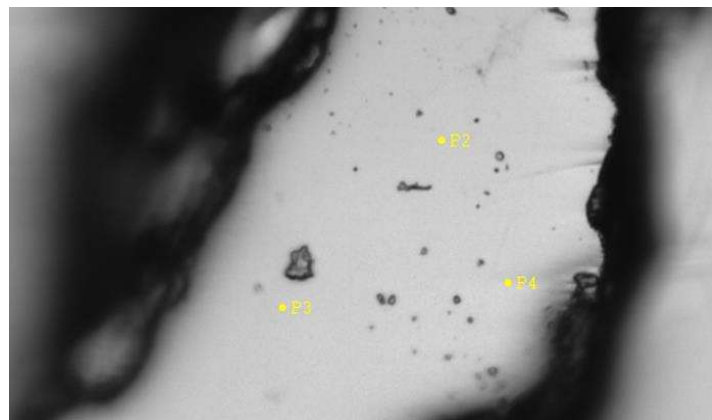


Fig.22 Showing nano indents created by hardness test of Ni-Ti interdiffusion zone.

S.No	Hardness Hv	Modulus Gpa
1	18.5	11.8
2	7.4	8.4
3	15.6	9.1

Table.5 Hardness and modulus at the three points in the Ti-Ni region

The average hardness and modulus in the Ni-Ti interdiffusion zone are 13.8 Hv and 9.7 Gpa, respectively. This hardness is significantly lower than the hardness of pure Ni and pure Ti. This could be due to cracking of the material due to thermal stress created during annealing.

9. Conclusion

This study used the diffusion couple technique to analyze hardness and modulus trends on alloys with different Fe, Ni, and Ti compositions. For the analysis, Ni-S.S. and Ni-S.S.-Ti alloy was prepared using the diffusion couple technique to analyze the composition versus hardness and modulus trend. The diffusion couple was found to be better than the conventional approach because it is a methodology through which numerous compositions of metal can be prepared using a single experiment. Also, the holding time and temperature parameters were optimized so as to get enough length of interdiffusion zone, so that the mechanical properties can be measured. The diffusion of Ni into S.S. and S.S. into Ni was found to be smooth when diffused for 100 hours. The hardness values were calculated through the nanoindentation test. These values in the Ni-rich region were found to be smaller than that for the S.S.-rich region. The length of the interdiffusion zone for Ni into S.S. was found to be around 150 micrometer. The diffusion of Ni into Ti and Ti into Ni was also found to be smoother for 100 hours. The hardness values at this region were found to be smaller than the hardness value for the individual metals, due to thermal stress created during annealing. The length of the interdiffusion zone for Ni into Ti. was found to be around 400 micrometers.

10. Scope for Future Work

The diffusion couple technique can be expanded to study various mechanical properties other than hardness and modulus, which change with composition. This technique can also be expanded to other elements whose alloys can be prepared in the laboratory with a large enough

diffusion zone. This technique can eliminate the use of the conventional one-alloy-at-a-time alloy approach. The study of a mechanical property obtained through this method can be useful to verify or check the usability of a material on any engineering application.

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